GARIMA MITTAL

garima.mittal@students.iiit.ac.in +91-6239749980 garima-mittal garimamittal Portfolio

Education

International Institute of Information Technology, Hyderabad **July 2023 – May 2027**
B.Tech in Electronics and Communications CGPA: 7.56 *Hyderabad, Telangana*

B.C.M Arya Model Sr. Sec. School **April 2009 – April 2023**
High School, 95% in +2 CBSE Boards *Ludhiana, Punjab*

Skills

Languages: C, C++, Python, TypeScript, JavaScript, SQL, MATLAB, Verilog, RISC-V Assembly.
Software Engineering: Data Structures & Algorithms, OOP, Operating Systems, Multithreading, Socket Programming, REST APIs, Microservices, Git, Linux, Kafka
Backend & Web: Node.js, Express.js, React.js, Next.js, MongoDB, PostgreSQL/Supabase, HTML/CSS
Tools: LATEX, PowerBI, Appium, WebdriverIO, BrowserStack, Claude Code
Soft Skills: Presentation skills, Team collaboration, Problem solving, Logical reasoning, Multitasking, Critical thinking.
Relevant Coursework: Statistical Methods in AI Operating System and Algorithms, Data Structures and Algorithms in C and C++, Cognitive Science and AI, Behavioral Research: Statistical Methods,Probability and Random Processes, Linear Algebra.

Experience

CHUBB (Software Intern) | ***Bengaluru, India*** ***May 2026 – July 2026***

- Built a mobile test automation framework using **Cucumber, TypeScript, Appium, WebdriverIO, and BrowserStack**, automating **100%** of business-provided Android test cases within **2** months through new automation methods and achieving **78%** coverage of priority iOS test cases, enabling automated testing across multiple applications within the domain.
- Extensively leveraged **Claude Code as an AI engineering agent** for framework design, implementation and debugging while reviewing and validating AI-generated solutions against application requirements and test behavior. Collaborated with Technical and Quality Engineering teams to integrate the framework into testing workflows.

Computer Vision–Based Spinal Motion Analysis | ***YOLO, MediaPipe, OpenCV, Python*** ***Jan 2026 – Present***

- Developing a computer vision pipeline to analyze human posture from **20-second exercise videos** across multiple bending and mobility movements. Tracking **four spinal reference points** (lower back, mid-back, neck, and shoulders) using YOLO detection and MediaPipe pose landmarks to estimate spinal flexion, joint angles, and posture alignment.

Projects

Distributed Network File System in C | ***C, Multithreading, Semaphores, Locks, Socket Programming***

- Designed a distributed file system using a client–server architecture with efficient metadata management, scalable storage operations, and redundancy handling.
- Improved system reliability by **30%** through fault-tolerant server mechanisms.

AI-powered Financial Dashboard | ***Next.js, Prisma, Supabase, Arcjet, Inngest, Gemini AI***

- Built a full-stack financial management platform with real-time transaction tracking, personalized budget analysis, and secure role-based access.
- Utilized dependencies such as **Arcjet** (security and rate limiting) and **Inngest** (event-driven workflows) to enhance reliability and scalability and integrated **Clerk** for authentication and access control.

IIIT-H Student Marketplace Platform | ***MongoDB, React, Express, Node.js***

- Developed and deployed a platform to facilitate buying and selling among IIIT-H students for seamless access and efficient transactions.
- Implemented user authentication, product listings, search functionality, and OTP verification for secure transactions.

Custom Linux-like Shell in C | ***C***

- Designed a modular shell supporting 15+ UNIX commands, including piping, I/O redirection, job control, and tab autocompletion.
- Implemented signal handling (`SIGINT`, `SIGTSTP`), background/foreground process management, and up/down arrow key navigation for command history.

Sentiment Analysis Web App | ***Flask, PyTorch, BERT, Transformers, scikit-learn, JavaScript***

- Engineered a full-stack web app for real-time **emotion detection** using fine-tuned **BERT**, with RoBERTa fallback for robust 6-class sentiment during college megathon.
- Executed emotion **intensity scoring (1–10)**, sentiment tracking, and dynamic **Matplotlib** visualizations in a responsive multi-page UI.

Extra-curricular activities

Teaching Assistant **August 2025 - Present**
IIIT Hyderabad *Hyderabad, Telangana*

- Served as a Teaching Assistant for **Linear Algebra** and **Systems Thinking** across **two semesters**, supporting undergraduate coursework through structured tutorial sessions.
- Evaluated courses supporting a combined strength of **120–400** students across two courses.

Design Head, Felicity Annual Fest **May 2025 - Present**
IIIT Hyderabad *Hyderabad, Telangana*

- Part of the design team for Felicity'25, designed posters for the annual fest.
- Designed 20+ promotional posters for Felicity '26, resulting in a 10% increase in student awareness of the annual festival.